

SAFE WORK METHOD STATEMENT (SWMS)

In order of priority:

- **ELIMINATION** – Remove hazard completely.
- **Substitution** – Replace object /process with less hazardous one.
- **Isolation** – Isolate the hazard (i.e. separate the hazard from the people).
- **Engineering** – Modify a process (i.e. redesign a workstation).
- **Administration** – Use procedures /systems to minimise the risk (i.e. Rotation of employees, implementation of a safe work method, training, etc).

LIKELIHOOD	RISK SCORE CALCULATOR				
Certain to Occur	Mod.	High	High	Extreme	Extreme
Very Likely	Mod.	Mod.	High	Very High	Extreme
Possible	Low	Mod.	High	Very High	Very High
Unlikely	Low	Mod.	Mod.	High	Very High
Rare	Low	Low	Low	Mod.	High
CONSEQUENCE	LOW	MINOR	MODERATE	FATAL	CATASTROPHIC

Residual Risk	Action Required
Low	Ok for now. Continue to monitor.
MODERATE	Determine if remedial action/treatment is required.
HIGH	Immediate decision required by site leadership. Identify and commence remedial action with appropriate priority.
VERY HIGH	ACT NOW. Requires immediate attention by site leadership. Address the risk without delay. Requires urgent attention and clear, timely action plans to reduce the risk before proceeding.
EXTREME	ACT NOW. Requires immediate attention by site leadership. Urgently address the risk. Requires immediate resolution with focused attention until resolved.

Identify task high risk workplace hazards - *Non routine work is work not documented in a Safe Operating Procedure (SOP)* or Work Instruction

Note: If machine isolation is required, workers must be competent in the relevant isolation procedure.

EXTREME & VERY HIGH RISK HAZARDS	Risk	MAIN RISKS / PERMITS REQUIRED
1. Entering a Confined Space.	E	- Heat exhaustion, engulfment, limited access/egress. - Asphyxiation – low O₂ or contaminants. (Confined Space Entry Permit required) - Explosive atmosphere, combustible dusts (Confined Space Entry/Hot Work Permits required)
2. Work in an area which may have a contaminated or flammable atmosphere.	E	- Fire / explosion from ignition source. (Hot Work Permit required) - Asphyxiation or poisoning from contaminated atmosphere. (Confined Space Permit)
3. Work with potential for a fall from 2m or work on a roof	E	Falling worker/object ≥2m requires SWMS.
4. Hot work required. (Oxy cutting, welding, soldering, grinding, etc).	E	- Fire or explosion from ignition source. (Hot Work Permit required) - Injury caused by hot surface, grinding wheel ejected pieces, welders "flash", etc. Fire inside Expanded Polystyrene (EPS) wall panels (Hot Work Permit required).
5. Hand dig >200mm OR ground penetration >200mm OR mechanical earthworks OR enter a trench >1.5 M deep (excavating is Construction work)	E	- Contact utilities – electricity, water, gas, communications. (Excavation Permit required) - Trench Collapse. (Excavation Permit required)
6. Non routine electrical, high voltage or 'live' work or work near electrical installations.	E	- High voltage is >33kV. 'Live' work is when electrical equipment is worked on energised – (Management Approval Required) - Electrical cabinets or MCC identified as Red
7. Non routine work involving mechanical lifting, lowering, pushing or pulling.	E	- High mechanical force damage / injury. - High risk work licence for use of cranes, mobile plant, scaffold, dogging, etc.
8. Using high risk tools with excess noise, kickback, cutting edge, high speed/pressure, ejected material, weight, awkward posture, explosive force.	E	- Injury from use of high risk powered or hand tools. - Electrocution, burn or electric shock. - Manual task injury from weight, forces, overexertion, repetition or poor work heights.
9. Demolish / alter load bearing / integral part of a structure 6m high OR use load shifting machinery on a suspended floor OR use explosives. Includes modify handrail / floors.	E	Being struck or crushed by parts of a structure or exposure to asbestos, lead, hazardous chemicals etc. (Demolition License and/or asbestos removal licence and Safe Work Plan required)
10. Work inside fences / barricades / enclosures where moving / rotating machinery is operating or hazards may exist.	V	- Serious bodily injury caused by contacting moving/unguarded plant or materials. - Machinery starting unexpectedly or electrocution, burn or electric shock.
11. Potential or stored energy which cannot be isolated (electrical, hydraulic, pneumatic, kinetic).	V	- Electrocution or entrapment. - Machinery entanglement / injury.
12. Work on or near a pressurized or enclosed circuit (water, pneumatic, hydraulic, hot oil, etc).	V	- Fracturing the mains or piping from contact – spills / leaks / emissions (environment). - Contacting temperature extremes.
13. Working in artificial temperature extremes or weather extremes. Hot, cold, wet, icy, lightning, high winds.	V	- Electrocution from using power tools / leads in wet conditions or lightning. - Heat stroke / Heat Exhaustion / Fatigue / Frost bite / Hypothermia / Influenza. - Slip / Trip / Fall Hazards.
14. Hazardous manual tasks or excessive noise or excessive vibration.	V	Personal injury/illness or hearing loss.
15. Using a hazardous chemical – chemicals, heavy metals, asbestos, Lead.	V	- Exposure to hazardous chemicals. - Spills / leaks (environmental).

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Key Steps in Logical Order		Hazards/Risks to Workers or Public	Risk Score	Control Measures	Residual Risk Score
1	Job planning and permits	Unplanned work. Conflicting activities. Inadequate controls.	Low	- Walk around to review scope of work and location of cracks. - Obtain hot work permit - Confirm competent personnel (welder, EWP operator, fire watch).	Low
2	Isolate plant and make safe	Unexpected start-up. Stored energy. Entanglement. Fire risk	High	- Lock out officer to isolate twin screws and airlock - Barricade area below EWP and weld zone. - Establish exclusion zone (falling objects, sparks). - Remove combustible materials. - Position fire extinguisher and fire watch.	Low
2	EWP inspection and setup	Falls from height. EWP failure. Collision.	High	- Pre-start inspection of EWP. - EWP operated by licensed operator. - Harness and lanyard fitted are connected.	Low
4	Access weld location and prepare surface	Falls from height. Sharp edges. Heat exposure.	High	- clean area around cracks (remove dust, residue). - Inspect for remaining hazards (heat, sharp edges). - Reconfirm hot work controls before welding starts.	Med
5	Perform welding works	Fire. Burns. Arc flash. Fumes. Falling sparks and slag.	High	- Perform welding in accordance with hot work permit. - Maintain fire watch during welding if permit re. - Control sparks and slag. - Monitor for heat transfer to adjacent equipment. - Stop work immediately if conditions change.	Med
6	Post-weld cooling and inspection	Residual heat. Smouldering material. Burns.	High	- Allow welds to cool. - Inspect weld quality. - Check surrounding areas for hot spots or smouldering material. - Maintain fire watch for required 30min post-work period.	Low

Attach additional pages if required

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All relevant workers to be consulted during this process and made aware of the control measures and all required permits identified.

AUTHORISATION			
Approved by	David Knight	Position	Plant manager
Signed		Date	

Name, Sign Off & Employer of Persons Performing Work		Start Time:	am/pm	Finish Time:	am/pm
Name	Sign Off	Employer			
1					
2					
3					
4					
5					

JOB COMPLETION				
1. Has equipment been modified/changed to affect how it operates?	☐ N (go to 3) ☐ Y (sign off by competent operator required)			
Has a competent operator approved it is safe to use? ☐ Y ☐ N	Competent Operator Name:		Competent Operator Signature:	
2. Work completed & area assessed safe before return to normal service? ☐ Y ☐ N <i>To be completed by the person conducting the task</i>	Name: Signature:			
3. Albioma Representative advised of & accepts changes: ☐ Y ☐ N	Name:		Signature:	

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